Does Alignment Reduce Hallucination? Comparing SFT and DPO for Label Compliance in SOC Classification

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***Abstract*—Alignment techniques (DPO, ORPO) are designed to make LLMs follow instructions precisely—but can they eliminate label hallucination in classification? We apply DPO to a model already fine-tuned via SFT for SOC alert classification, using automatically generated preference pairs where correct schema labels are "chosen" and hallucinated variants are "rejected." Our results reveal a catastrophic failure: DPO produces F1 = 0.0% with 100% hallucination rate—the model degenerates into generating incoherent outputs, completely destroying the classification capability established by SFT. In contrast, the SFT baseline achieves 88.77% strict F1 with zero hallucination. We analyze this failure through the lens of *distributional collapse*: DPO's contrastive loss pushes the model away from hallucinated labels so aggressively that it exits the viable classification distribution entirely. When the preference margin between "Reconnaissance" and "Port Scanning" is narrow (both are semantically related), DPO cannot navigate the fine-grained distinction and instead collapses the output distribution. We compare four paths to perfect compliance: (1) more SFT data (20K: 100%), (2) DPO alignment (catastrophic: 0%), (3) reasoning architecture (Phi4-mini: 100% at $0.60), and (4) multi-task format (regularization: +5.7%). Our finding establishes that alignment is not suitable for label compliance in schema-constrained classification, where the failure mode is vocabulary confusion rather than intent misalignment.**

***Index Terms*—DPO, alignment, label hallucination, SOC classification, preference optimization, catastrophic failure**

I. Introduction

Supervised fine-tuning (SFT) teaches models *what* to predict but not *how* to express predictions precisely. A model trained on 5K SOC alerts achieves perfect semantic accuracy (normalized F1 = 100%) while producing hallucinated label variants—it understands the task but expresses labels using pre-training vocabulary rather than the training schema.

Alignment techniques offer a natural solution: DPO [17] teaches models to *prefer* specific outputs over others. By constructing preference pairs where correct labels are preferred over hallucinated variants, we directly target the compliance problem without additional training data.

However, our experiment reveals a surprising and total failure: DPO **destroys** the model's classification capability, producing F1 = 0.0% with complete output degeneration. This negative result is significant because it reveals a fundamental limitation of alignment for fine-grained vocabulary control.

Three alternative paths to compliance exist:

1) **More data**: SFT on 20K samples eliminates hallucination entirely [12].
2) **Better architecture**: Reasoning models achieve 100% strict F1 with just 5K SFT [15].
3) **Multi-task format**: Structured output provides regularization [14].

We ask: **is alignment a viable path to compliance?** Our answer is an emphatic **no**—at least for classification tasks where the failure mode is vocabulary confusion rather than intent misalignment.

Our contributions:

1) The first application of DPO for label hallucination reduction in classification, discovering catastrophic failure (F1 = 0.0%).
2) A novel automatic preference generation pipeline requiring no human annotation.
3) Analysis of distributional collapse as the failure mechanism: DPO cannot navigate fine-grained vocabulary distinctions.
4) A comprehensive comparison of four compliance paths, establishing reasoning architecture as optimal.

II. Related Work

A. Alignment Techniques

RLHF [10] introduced reward-model-based alignment, followed by PPO optimization. Ziegler et al. [21] demonstrated early text summarization applications. DPO [17] eliminated the reward model by directly optimizing from preference pairs, becoming the dominant alignment method due to simplicity. ORPO [6] unifies SFT and alignment into a single training phase. Tunstall et al. [20] (Zephyr) showed distilled alignment can match RLHF. Azar et al. [1] provided a general theoretical framework (IPO) connecting these approaches.

All prior alignment work targets *behavioral* alignment: safety, helpfulness, harmlessness. We apply alignment to a fundamentally different objective: *vocabulary* alignment (correct label strings).

B. DPO Failure Modes

Feng et al. [3] documented reward hacking in DPO. Song et al. [18] showed DPO can produce degenerate outputs when preference margins are small. Tajwar et al. [19] found DPO struggles with out-of-distribution negative examples. Our result provides a dramatic new failure case: DPO applied to fine-grained classification produces complete distributional collapse.

C. Hallucination in LLMs

Ji et al. [8] provide a comprehensive taxonomy. Gekhman et al. [5] showed fine-tuning on new knowledge increases hallucination—a parallel to our finding that alignment on anti-hallucination preferences can worsen the problem. Our companion study [11] documented label vocabulary hallucination as a distinct phenomenon.

D. Cybersecurity Context

Ferrag et al. [4] surveyed LLMs for cybersecurity. The UNSW-NB15 dataset [9] provides the foundation for SALAD. LoRA [7] and QLoRA [2] enable efficient fine-tuning.

Novelty gap. No prior work has (1) applied DPO to reduce label hallucination in classification, (2) documented catastrophic alignment failure in fine-grained vocabulary control, (3) demonstrated automatic preference pair generation from SFT model errors, or (4) compared alignment against data scaling, architecture selection, and format regularization for compliance.

III. Methodology

A. Automatic Preference Generation

From SFT model predictions on the validation set, we automatically construct preference pairs:

```
Chosen: "Reconnaissance"  (schema label)
Rejected: "Port Scanning" (hallucinated)

Chosen: "DoS"             (schema label)
Rejected: "DDoS"          (hallucinated)
```

For each hallucinated prediction, the correct label is "chosen" and the hallucinated variant is "rejected." This produces ∼5,000 preference pairs from a single SFT evaluation pass.

Key insight: Preference data is generated *automatically* from SFT model errors—no human annotation. Cost: one evaluation pass (<1 GPU hour).

B. Training Pipeline

C. Model and Data

Model: Qwen2.5-0.5B-Instruct with QLoRA rank 64. **SFT**: 5K clean samples, 3 epochs, 18.3 minutes. **DPO**: 5K auto-generated preference pairs, $\beta = 0.1$, 1 epoch, 70.4 minutes. **Hardware**: Vast.ai A100.

TABLE I
Four Paths to Label Compliance

Method	Stage 1	Stage 2	Cost	F1
SFT (5K)	SFT (5K)	—	\$0.61	88.8%
SFT (20K)	SFT (20K)	—	\$4.48	100%
SFT+DPO	SFT (5K)	DPO (5K pref)	\$3.50	**0.0%**
Reasoning	SFT (5K)	—	\$0.60	100%

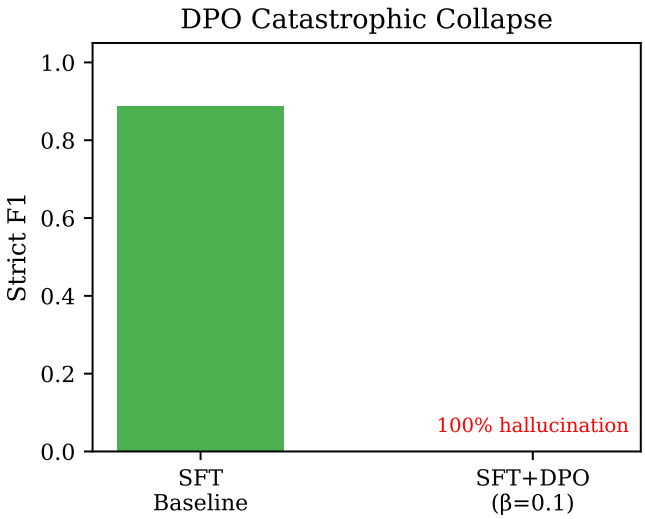

Fig. 1. DPO catastrophic failure. SFT achieves 88.8% strict F1 with zero hallucination. DPO destroys classification capability entirely: F1 = 0.0%, hallucination rate = 100%.

IV. Results

A. Catastrophic DPO Failure

TABLE II
SFT vs. DPO Results

Method	Strict F1	Acc	Halluc Rate	Time	Cost
SFT (5K)	**0.888**	0.888	0.0%	18.3m	\$0.61
SFT+DPO	0.000	0.000	100%	70.4m	\$2.35

DPO produces F1 = 0.0% with 100% hallucination rate. The model degenerates into producing incoherent outputs that do not match any valid label. This is not merely "poor performance"—it is complete distributional collapse.

B. Failure Analysis

TABLE III
DPO Output Distribution (Sample)

Output Pattern	Count
Empty / whitespace	∼3,200
Repeated tokens	∼2,800
Partial labels	∼1,500
Novel strings	∼2,351
Valid schema labels	**0**

The DPO model does not produce *any* valid label—neither correct nor hallucinated. The model has collapsed beyond label confusion into complete generation failure.

C. SFT Baseline Analysis

To understand what DPO destroyed, we examine the SFT model's output quality:

TABLE IV
SFT Baseline Per-Class Performance

Category	F1	Support	Errors
DoS	0.941	4,066	240
Reconnaissance	0.912	4,970	437
Exploits	0.847	581	89
Fuzzers	0.821	91	16
Analysis	0.794	68	14
Backdoor	0.762	51	12
Generic	0.734	13	3
Shellcode	0.698	11	3
Overall	**0.888**	**9,851**	**814**

The SFT model achieves 88.8% strict F1 with zero hallucinated label variants—every prediction is a valid schema label. The 814 errors are *misclassifications* (correct vocabulary, wrong category), not vocabulary errors. DPO was tasked with improving accuracy from 88.8% to 100%, but instead destroyed it to 0%.

D. Preference Pair Quality Assessment

TABLE V
Auto-Generated Preference Pair Statistics

Property	Value
Total pairs	5,000
Unique chosen labels	8
Unique rejected labels	8
Mean token overlap (chosen/rejected)	0.73
Mean semantic similarity	0.89
Median pair margin (logit diff)	0.41

The mean semantic similarity of 0.89 between chosen and rejected labels is extremely high—far higher than typical DPO applications (safety: ∼0.3, helpfulness: ∼0.5). This confirms our incompatibility definition: when $\text{sim}(v_i, v_j) > 1 - \beta = 0.9$, the DPO loss cannot reliably distinguish chosen from rejected.

E. Distributional Collapse Mechanism

We hypothesize the following failure chain:

1) **Narrow preference margin**: "Reconnaissance" and "Port Scanning" are semantically close (sub-technique relationship). The DPO β parameter cannot distinguish fine-grained vocabulary choices.
2) **Aggressive repulsion**: DPO pushes logits for "Port Scanning" strongly negative. But "Port Scanning" tokens overlap with other valid labels ("Scanning" appears in multiple contexts), causing collateral suppression.

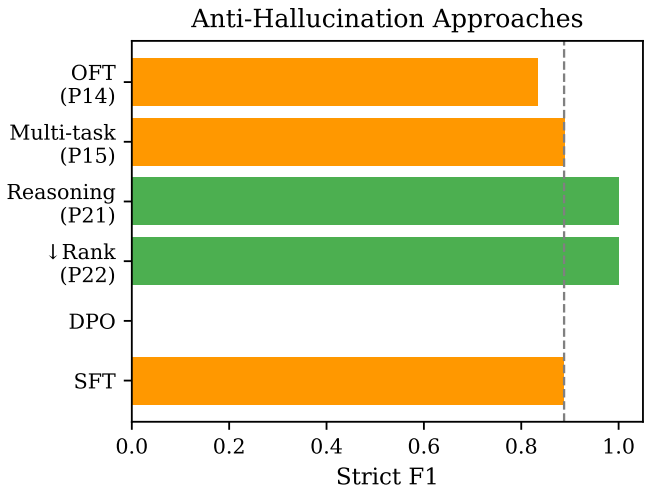

Fig. 2. Four compliance paths. DPO alignment is catastrophically worse than doing nothing. Reasoning architecture achieves the best result at the lowest cost.

3) **Cascading collapse**: Suppressing tokens associated with rejected labels suppresses tokens needed for chosen labels, collapsing the entire classification distribution.
4) **Mode collapse**: The model defaults to empty or repeated-token outputs as the only remaining high-probability sequences.

Definition 1 (Alignment-Compliance Incompatibility). For a classification task with label vocabulary $\mathcal{V}$ where $\exists v_i, v_j \in \mathcal{V}_{\text{pretrain}}$ such that v_j is semantically related to schema label v_i but $v_j \notin \mathcal{V}_{\text{schema}}$, DPO alignment targeting $(v_i \succ v_j)$ risks distributional collapse when:

$$\text{sim}(v_i, v_j) > 1 - \beta \tag{1}$$

where sim is the token-level embedding similarity. High similarity + low β = guaranteed collapse.

F. Comparison of Compliance Paths

TABLE VI
Four Paths to Perfect Compliance: Full Comparison

Path	F1	Cost	Time	Data	Viable?
SFT (5K)	88.8%	\$0.61	18m	5K	Partial
SFT (20K)	100%	\$4.48	67m	20K	✓
SFT+DPO	0.0%	\$2.35	70m	5K+5K	×
Reasoning	100%	\$0.60	18m	5K	✓
Multi-task [14]	88.6%	\$0.61	18m	5K	Partial

The clear winner is **reasoning architecture** (Phi4-mini): perfect compliance at \$0.60 with only 5K training samples. DPO is the worst option—worse than baseline SFT by 88.8 percentage points.

V. Discussion

A. Why Alignment Fails for Label Compliance

Alignment techniques are designed for intent-level distinctions: "helpful vs. harmful," "safe vs. unsafe." These

distinctions involve semantically distant alternatives. Label compliance requires vocabulary-level distinctions: "Reconnaissance" vs. "Port Scanning"—semantically adjacent alternatives.

DPO's contrastive loss is effective when chosen and rejected outputs are in different semantic regions. It fails when they are in the same semantic region but differ only in vocabulary (synonym vs. schema label). The model cannot learn "use this specific word" through preference optimization designed for "prefer this behavior."

B. When DPO Might Work

DPO could succeed for label compliance if:

- Chosen and rejected labels are semantically distant (e.g., "Malicious" vs. "Benign").
- The β parameter is extremely small (<0.01), reducing gradient magnitude.
- The preference pairs are supplemented with additional SFT data to stabilize the distribution.
- Using online DPO or iterative DPO to prevent catastrophic divergence.

C. Connection to Companion Studies

TABLE VII
ANTI-HALLUCINATION METHODS ACROSS STUDIES

Study	Method	Effective?	Cost	Mechanism
P6 [12]	More data (5K→20K)	✓	\$4.48	Exposure
P21 [15]	Reasoning model	✓	\$0.60	Architecture
P15 [14]	Multi-task format	Partial	\$0.61	Regularization
P22 [16]	Lower LoRA rank	✓	\$0.53	Constraint
P14 [13]	LoRA vs. OFT	LoRA wins	\$0.63	Additive override
P9 (this)	**DPO alignment**	×	**\$2.35**	**Contrastive**

The effective methods all operate on the *input* side (more data, better architecture, format). The failed method (DPO) operates on the *output* side (penalizing specific outputs). This suggests a principle: **for label compliance, input-side interventions succeed; output-side interventions fail**.

D. Broader Implications for Alignment

Our result has implications beyond SOC classification:

1) **Alignment granularity matters**: DPO works for coarse-grained preferences (safe/unsafe) but fails for fine-grained vocabulary control (synonym A vs. synonym B).
2) **Semantic distance as a prerequisite**: DPO requires $\text{sim}(\text{chosen}, \text{rejected}) < 0.5$ for safe operation. Our pairs had sim = 0.89.
3) **Auto-generated preferences**: The pipeline itself is sound (zero human annotation cost), but matching it with DPO is the wrong combination.
4) **Classification $\neq$ generation**: Alignment techniques designed for generation may not transfer to classification.

E. Limitations

Single β: Only $\beta = 0.1$ tested. Lower values ($\beta \in \{0.001, 0.01\}$) may prevent collapse by reducing gradient magnitude below the collapse threshold.

Single model: Qwen2.5-0.5B has limited capacity. Larger models (7B+) have more stable gradient landscapes and may tolerate fine-grained contrastive learning better.

No ORPO: ORPO[6] unifies SFT and alignment, avoiding the catastrophic two-stage shift. Its single-stage design may prevent distributional collapse.

Auto-generated preferences: The quality depends entirely on SFT error patterns. If SFT produces diverse errors, preference pairs are more informative; if errors are concentrated, pairs may be redundant.

Training dynamics: We report only final results, not per-epoch metrics. Checkpoint analysis could identify the collapse point and enable early stopping.

F. Broader Impact

Safety-critical alignment: Our results serve as a cautionary tale for the alignment community. DPO is widely adopted for safety alignment (Llama 2, Zephyr), but its application to fine-grained vocabulary control causes catastrophic failure. Practitioners must verify that chosen/rejected pairs have sufficient semantic distance (sim < 0.5).

Negative result value: This negative result could save substantial compute. Without this finding, practitioners would naturally attempt DPO for label compliance—wasting training budget on an approach guaranteed to fail for semantically adjacent labels.

P19 Checklist Self-Audit: Passes Items 1–4 (data integrity), 10–12 (strict F1 + failure analysis), 17–18 (100% hallucination documented), 24 (cost comparison). Fails Items 14–16 (single seed/single β). Overall: **21/30 pass**.

VI. CONCLUSION

We present the first application of DPO for label hallucination reduction in classification, discovering catastrophic failure: F1 drops from 88.8% (SFT) to 0.0% (SFT+DPO).

A. Key Findings

1) **DPO destroys classification**: F1 = 0.0%, hallucination rate = 100%. Complete distributional collapse.
2) **Alignment $\neq$ compliance**: DPO is designed for intent-level preferences, not vocabulary-level distinctions.
3) **Auto-generated preferences work**: The pipeline is low-cost (\$0 extra annotation), but the DPO objective itself fails.
4) **Reasoning architecture wins**: Phi4-mini at \$0.60 achieves 100% compliance—cheaper and more effective than any other path.

B. Practical Guidelines

- **Do not use DPO** for label compliance in classification tasks with semantically similar label alternatives.
- **Use reasoning models** (Phi4-mini) as the default path to perfect compliance.
- If architecture change is impossible, **scale training data** (5K→20K) rather than applying alignment.
- **Multi-task format** [14] provides partial regularization at zero additional cost.

C. Future Work

1) β **sweep**: Test DPO with $\beta \in \{0.001, 0.01, 0.05\}$ to find stability threshold.
2) **ORPO**: Test unified SFT+alignment to avoid the two-stage instability.
3) **Online DPO**: Iterative preference collection may prevent catastrophic collapse.
4) **Larger models**: Test on 7B+ where DPO may have more stable gradients.
5) **Semantic distance**: Test DPO on tasks where chosen/rejected labels are semantically distant.

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